

Predicting Polycystic Ovary Syndrome with Machine Learning

A plain-language report on what we built, what we found,
and what we could not do

Group 5

Fatema Ferdous · 0152410052

Wafa Haque · 0152420023

Khondoker Sazzad Sunfi · 0152310002

The short version. We built a system that predicts PCOS from routine medical measurements, and it works well — about **93% accurate**. But the more useful part of this project is what we found while checking our own work: several published papers report 98–99% accuracy on this exact dataset, and we show that a large part of such a score can come from how you test rather than how good your model is.

Machine Learning course project

1. The problem, in plain terms

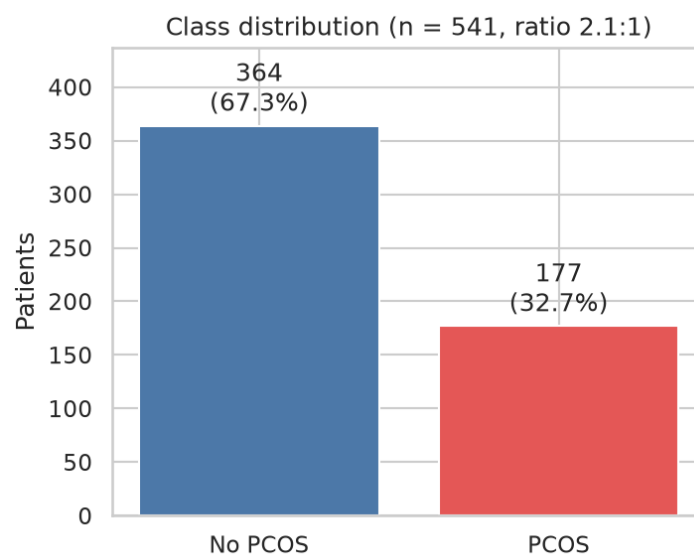
Polycystic Ovary Syndrome (PCOS) is a hormonal condition affecting roughly 8–13% of women of reproductive age. It causes irregular periods, weight gain, excess hair growth and skin changes, and if it goes undetected it raises the risk of infertility, type 2 diabetes and heart disease.

The difficulty is that these symptoms are common and overlap with other conditions. Proper diagnosis needs blood tests and an ultrasound scan, which are not always quick, cheap or available — especially in areas with limited healthcare resources.

Our question: can a computer program look at a woman's routine measurements and flag whether she is likely to have PCOS, so that a doctor knows who to investigate further?

2. The data

We used a public dataset of **541 women** from 10 hospitals in Kerala, India, with **41 measurements** recorded for each person — age, weight, blood pressure, hormone levels, symptoms, and ultrasound findings.



Of the 541 women, 177 have PCOS and 364 do not — about two healthy women for every patient. This imbalance matters: a lazy program could score 67% just by guessing 'no PCOS' every single time.

A dataset with mistakes in it

Before doing anything clever, we had to clean the data — and it needed more cleaning than expected. Among other problems we found **13 physically impossible values**:

- A blood pressure of **12/80** and another of **120/8** — clearly someone dropped a digit from 120/80.
- Pulse rates of **13 and 18 beats per minute**. A person with that pulse is not sitting in a clinic answering questions.

- A hormone reading of **5052** where normal is under 25 — a decimal point in the wrong place.

We **blanked** these rather than guessing what they should have been, and let the program fill them in the same way it fills any other missing value. Guessing '120/80' would look tidier but would be inventing data.

Worth noting: four of the five published papers we reviewed use this same dataset, and none of them mention these errors. They almost certainly trained on a pulse rate of 13 bpm without noticing.

3. What we built

We built an automated pipeline — a fixed sequence of steps that runs the same way every time:

Step	What it does	Why
1. Clean	Fix the broken values described above	Rubbish in, rubbish out
2. Fill gaps	Estimate the handful of missing numbers	Models cannot handle blanks
3. Rescale	Put every measurement on a comparable scale	Otherwise 'weight in kg' outshouts 'number of symptoms'
4. Choose features	Keep the most informative measurements	Fewer inputs, less chance of memorising noise
5. Balance	Even out the 2:1 imbalance	So PCOS cases are not ignored
6. Predict	One of nine different algorithms	No single one is obviously best

The critical detail is **where** steps 2–5 happen. They all sit inside the pipeline, so they are recalculated from scratch every time the model is tested on a new group of patients. That sounds like a technicality. It is the single most important decision in the project, and Section 4 explains why.

How we tested it honestly

We set aside **20% of the women at the very start** and did not look at them again until the end. Everything — which features to use, which algorithm to pick — was decided using only the other 80%.

Think of it as an exam. The training data is the textbook; the held-out 20% is the exam paper, kept in a sealed envelope. If you peek at the exam while revising, your score stops meaning anything.

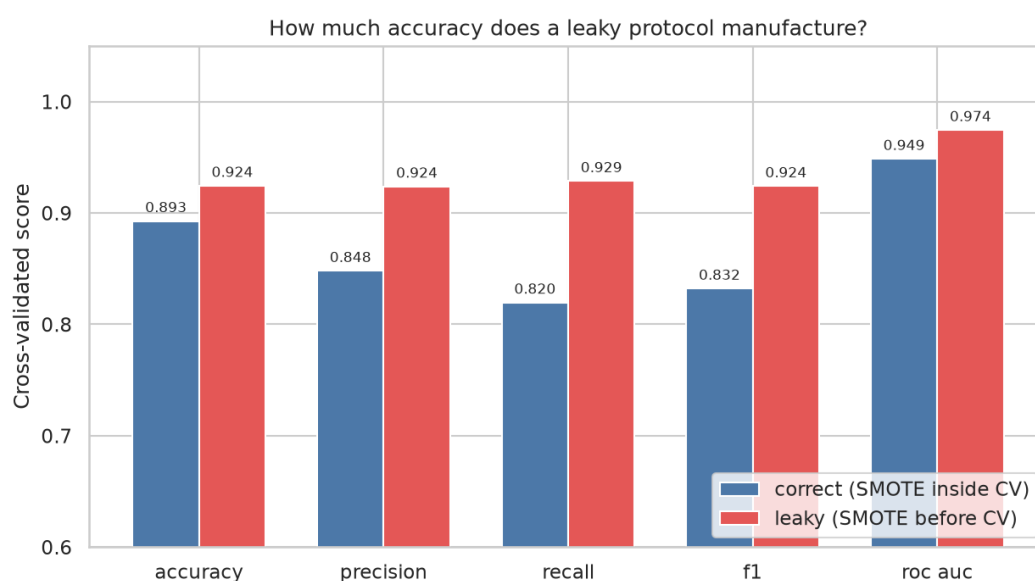
4. Finding 1: how to accidentally cheat

Published papers using this dataset report accuracies of 98–99%. We reach about 93%. Rather than argue about it, we **measured** where such a gap can come from.

Remember step 5, balancing the data. It works by inventing extra synthetic PCOS patients, each built by blending real ones. Now consider the order of operations:

- **The right way** — split the patients first, then invent synthetic ones using only the training group.
- **The tempting way** — invent synthetic patients first, then split. Now a synthetic patient built from a training woman can land in the exam envelope. The model is partly being tested on copies of what it studied.

We ran both, with the same model, the same data and the same patients:



Same model, same data, same split. The only difference is the order of two steps.

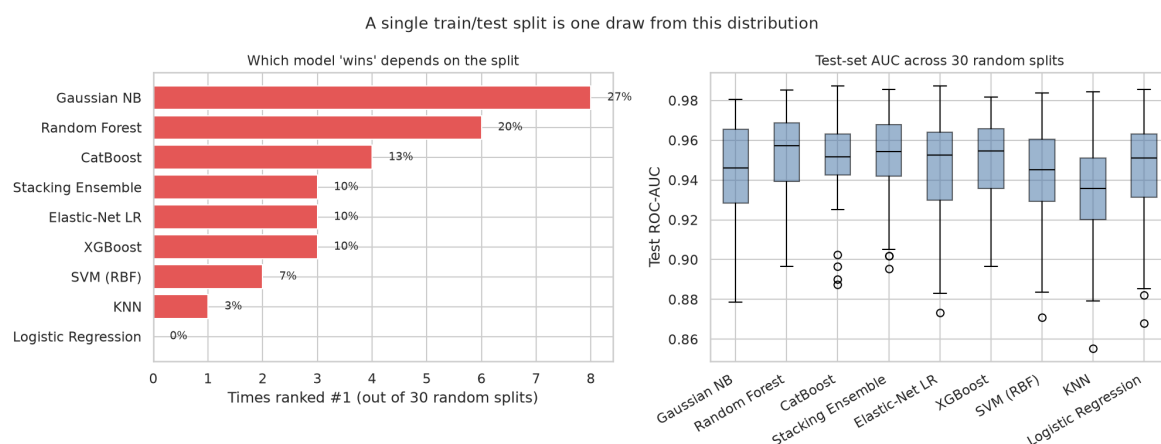
Doing it the tempting way inflates accuracy by **3.2%** and the F1 score by **9.2%** — without the model getting any better at all. A 98–99% headline on this dataset is **reachable through testing method alone**.

This does not prove the other papers made this mistake. Most simply do not describe their method in enough detail to tell — which is itself the finding. A number you cannot check is a number you cannot trust.

5. Finding 2: there is no 'best' algorithm here

We compared **nine** different algorithms, from simple logistic regression to a stacked ensemble of several models. Papers routinely announce a winner. We tested whether a winner even exists.

We re-shuffled which patients go into training and which into the exam envelope **30 times**, and recorded which algorithm came first each time:



Each algorithm's score swings by 8–13 percentage points depending purely on which patients happened to land in the test set.

8 of the 9 algorithms won at least once. The most frequent winner took only 27% of the rounds — and it was Gaussian Naive Bayes, one of the simplest methods available. A paper that declares a best model from a single split is reporting a coin flip.

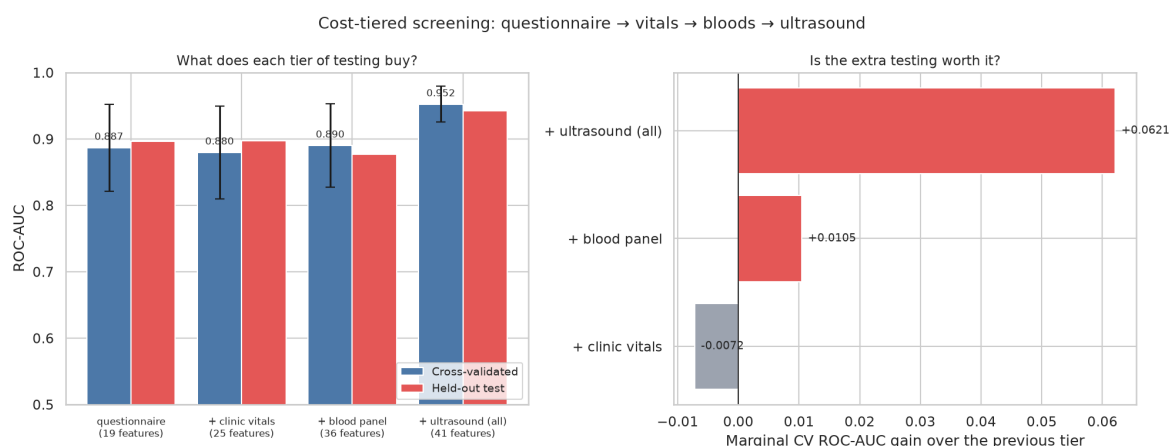
This is not a failure of the models. It is a consequence of having 541 patients. With only 36 PCOS cases in the exam envelope, three women changing sides moves the accuracy by several points.

6. Finding 3: you may not need the expensive tests

Our introduction argued that PCOS testing is often hard to access. So we asked a question none of the reviewed papers ask: **what does each level of testing actually buy you?**

We grouped the measurements by how much they cost to obtain, and trained a separate model on each level.

What you need	Model quality (AUC)	Gained by adding this step
A form, a scale, a tape measure	0.887	—
+ a nurse for 5 minutes	0.880	-0.0072
+ blood draw and lab tests	0.890	+0.0105
+ ultrasound scan and a sonographer	0.952	+0.0621



Left: how good the model is at each level of testing. Right: what each extra step adds. Only the ultrasound bar is large.

A questionnaire alone — no doctor, no laboratory, no ultrasound — reaches 0.887, which is **93% of the full model**. The entire blood panel (nine lab tests and a needle) adds +0.0105. Only the ultrasound genuinely pays for itself, at +0.0621.

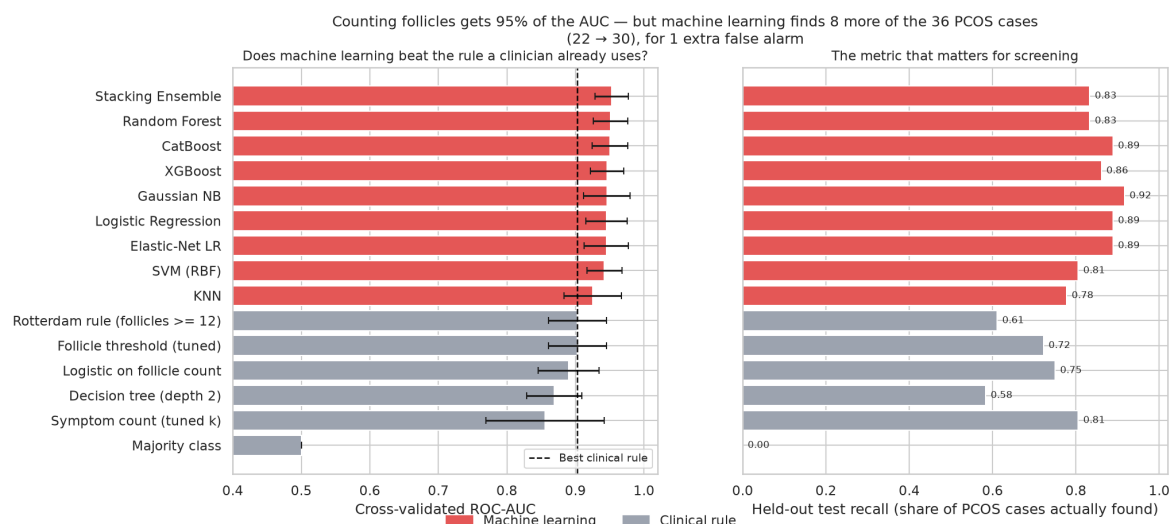
The practical recommendation: in a clinic that cannot afford everything, run the questionnaire, skip the blood tests, and spend the budget on ultrasound access for the women the questionnaire flags.

7. Finding 4: does any of this beat what doctors already do?

This is the question we think matters most, and none of the five papers we reviewed asks it.

Doctors already have a rule. The Rotterdam criteria, agreed in 2003, say: if an ovary shows **12 or more follicles**, that counts toward a PCOS diagnosis. That is not machine learning. It is counting, with a threshold.

So we ran that rule through exactly the same tests as our nine algorithms.



Red = machine learning, grey = simple rules. Left: overall quality. Right: what fraction of actual PCOS cases each approach finds.

The uncomfortable half of the answer

Just counting follicles scores **0.903**. Our best model, after nine algorithms, data balancing, feature selection and an automated architecture search, scores **0.953**.

A rule from 2003 with **nothing learned from data at all** achieves **95%** of what all our machine learning achieves. Most of the signal in this dataset is simply the follicle count.

The half that justifies the work

But overall scores hide something important. The old rule is very good at correctly clearing healthy women, and much worse at the job it exists to do — finding patients:

Approach	PCOS cases found (of 36)	False alarms (of 73)
Counting follicles (the 2003 rule)	22	1
Our machine learning model	30	2

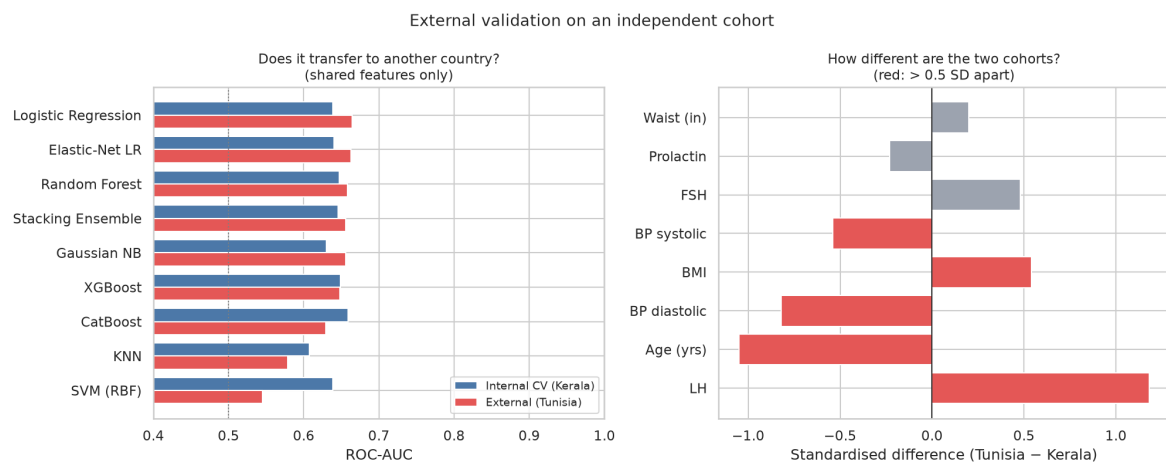
The model finds **8 more women with PCOS** and raises **1 extra false alarm**. The old rule misses 39% of the patients it is meant to catch; our model misses 17%. A false alarm costs one extra scan. A missed case costs years of undiagnosed illness.

So the honest conclusion is neither 'machine learning works' nor 'machine learning is pointless'. Almost all the information is in the follicle count, which a doctor can already read off a scan. What the model adds is a better decision about where to draw the line — and for a screening tool, that is exactly the job.

8. Finding 5: does it work on different women?

Every paper we reviewed was criticised in our literature review for testing only on the population it was built from. We tried to do better.

We found a genuinely independent study — **88 women from a hospital in Sfax, Tunisia** — and tested our model on them. Different country, younger patients, heavier on average, and a different balance of cases. We checked carefully that it was not simply a re-upload of our own training data, which many public 'PCOS datasets' turn out to be.



Left: performance at home (blue) versus abroad (red). Right: how different the two groups of women actually are.

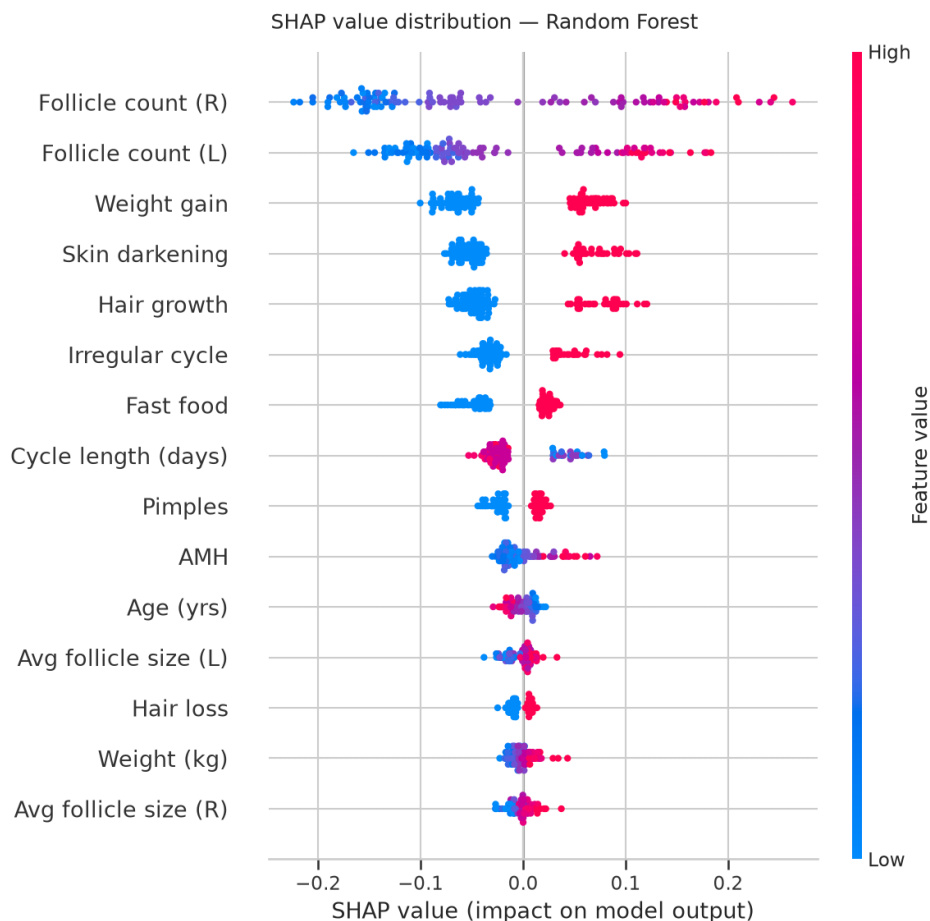
The good news: performance barely moved — an average change of -0.006. What the model learned is not specific to Kerala.

The catch: the Tunisian study recorded only 8 of our 41 measurements, and it did not record follicle counts or symptoms — the very things that carry the signal. So we could only test a weakened version of our model.

This turns into a stronger criticism of the field than we expected. The five papers did not merely neglect to test on other populations — **no public dataset exists that would let anyone do it**. What the field needs is not better algorithms but compatible data collection across hospitals.

9. What the model actually pays attention to

A prediction nobody can explain is not much use to a doctor. We used a technique called SHAP, which breaks each individual prediction down into how much each measurement pushed it one way or the other.



Each dot is one patient. Dots to the right pushed the prediction toward PCOS. Red means a high value of that measurement, blue means low.

The model's priorities, in order, are: **Follicle count (R), Follicle count (L), Weight gain, Skin darkening, Hair growth.**

That is reassuring, because it matches the three things doctors already look for: ovary appearance, signs of excess male hormones, and irregular cycles. The model was never told the medical criteria — it recovered them from data.

But there is a catch we should state plainly. Follicle count is both the model's top clue and one of the three criteria doctors use to record the diagnosis in the first place. So the model is partly learning the definition of the answer. Our questionnaire-only model is arguably more interesting, because none of its inputs are diagnostic criteria.

10. What we could not do

A report that only lists successes is not a scientific report. These are the real limits of this work.

- **Our numbers are less precise than they look.** With only 36 PCOS cases in the test set, the fraction of patients we correctly find could plausibly be anywhere from

70% to 94%. Every single-split number published on this dataset has that same imprecision; ours is just the one that says so.

- **The diagnosis itself may be circular.** The doctors who recorded the measurements also made the diagnosis, and were not blinded to it. Symptoms like 'weight gain' may have been asked about because PCOS was already suspected.
- **We do not know who these women are.** The dataset publishes no collection dates, no eligibility rules, and no treatment history. A third of them have PCOS, against 8–13% in the general population, so these are clearly women who already had reason to attend a hospital.
- **No ultrasound images.** We have follicle counts taken from scans, but not the scans themselves.
- **Our external test was partial.** As explained in Section 8, only a weakened version of the model could be tested abroad.
- **This is a coursework project, not a medical device.** Diagnosing PCOS requires a clinician.

We also applied a professional reporting standard (TRIPOD+AI) to our own work and graded ourselves against it. The verdict: our **analysis** is sound and we can defend it in detail; the **data** carries real risks we cannot fix. No amount of careful modelling repairs a diagnosis recorded by someone who already knew the answer.

11. Summary

What we built and what it scores:

Measure	Result	In plain terms
Accuracy	92.7%	Correct on this many patients overall
Sensitivity	83.3%	Of women who have PCOS, this many are found
Specificity	97.3%	Of healthy women, this many are correctly cleared
AUC	0.945	Overall ability to rank patients by risk (1.0 is perfect)

The five things worth remembering

- A flawed testing method can manufacture **3.2% extra accuracy** out of nothing. Published scores are only as trustworthy as the method behind them.
- **8 of 9 algorithms** won at least one round of testing. Arguing about which algorithm is best on data this size is arguing about noise.
- A **free questionnaire** gets you 93% of the way. The expensive blood panel adds almost nothing.
- A **2003 clinical rule** with nothing learned from data achieves 95% of our model's score — but our model finds 8 more patients.

- **Nobody can properly test PCOS models across populations**, because the necessary data has not been collected in a compatible way.

Our overall conclusion. Machine learning can support PCOS screening, and it does genuinely find patients a simple rule misses. But on this dataset the interesting variable was never the algorithm — it was the testing method. We think reporting a defensible 93% with its limitations stated is worth more than another 99% that nobody can check.

Full source code, all 30 figures, 31 result tables, 64 automated tests and a step-by-step notebook are in the project repository. Every number in this report is read directly from the pipeline's output when the report is built, so it cannot drift out of date.